

Clean-up operation

In 1989, the *Exxon Valdez* oil tanker struck a reef off Alaska, spilling more than 11 million gallons (41 million liters) of oil in the Pacific Ocean. Skimmers were used to help with the cleanup, removing the layer of oil that floated on the surface of the water. Each had a rope that was dragged across the water's surface to collect oil and pull it back to the boat to be squeezed out. Despite rigorous skimming, some oil still lingered in the water, harming wildlife and causing pollution for decades.

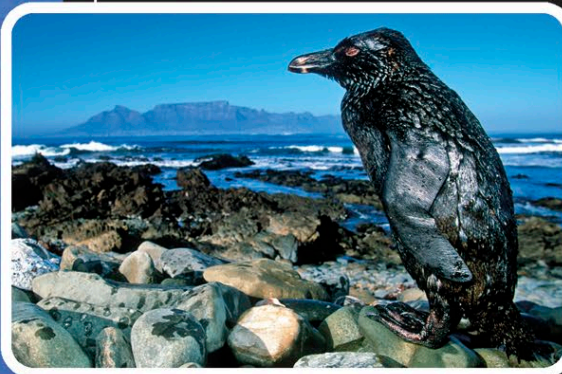
CLEANING UP OIL SPILLS

OIL SKIMMERS

Oil spills from tankers and oil rigs must be removed from the ocean and seas quickly, because they are likely to spread the longer they stay. It is vital to clean up oil spills to minimize their damage to wildlife and the environment.

Oil is also a fire hazard and can poison drinking water. Spills are tackled using devices called oil skimmers that remove the oil floating on the water's surface. Oil sticks to the material in the skimmers and is easily removed.

Some skimmers only clear away the oil, while others collect it for future use.



ENVIRONMENTAL COST

Spilled oil can stick to the coats of marine creatures and seabirds, making it difficult for them to swim, fly, or breed. It also poisons their food sources, which leads to toxins building up in the food chain, causing long-term problems.

DENSITY

Skimmers work because oil is less dense than water and forms a separate layer on the surface. The density of a substance is how much matter is packed into the space it takes up. The particles in water are more closely packed than those in oil, so water is more dense than oil. Liquids of different densities often form separate layers when mixed together. As a result, oil floats on the surface of water.

